# Unit 4: Differential Equations - Study Notes

## Topic 4.1: Concept, Definition, Order, and Degree

**Definition:** A differential equation is a mathematical equation that connects a variable, its value, and its rate of change (derivative) together. Order is the highest number of times a derivative has been taken in the equation. Degree is the mathematical power of that highest-order derivative.

**3 Key Features:**

• Dynamic Relationship: It describes how a system updates itself based on its current state.

• Order Identification: In an equation with both dy/dx and d^2y/dx^2, the order is 2 because the highest derivative is a second derivative.

• Degree Rules: The degree is the exponent of the highest derivative term, provided all derivatives are written as whole powers.

**3 Major Uses:**

• System Modeling: Creating mathematical representations of physical systems that change over time.

• Fluctuation Analysis: Tracking how fields like temperature or voltage shift across a physical space.

• Mechanical Vibrations: Labeling equations that define how springs bounce or strings vibrate.

**Real-Life Example:** Think of a cooling cup of tea. The rate at which the tea cools down depends directly on the difference between its current temperature and the room temperature. This relationship is written as a first-order differential equation.

## Topic 4.2: Variable Separable Method

**Definition:** The Variable Separable Method is a fundamental technique used to solve simple first-order differential equations. It works by moving all terms containing the variable x to one side of the equation and all terms containing the variable y to the opposite side so they can be integrated independently.

**3 Key Features:**

• Algebraic Sorting: It transforms a mixed derivative equation into a split form, such as f(y) dy = g(x) dx.

• Direct Integration: Once the variables are completely separated, you solve the equation by taking the regular integral of both sides.

• Explicit Solutions: It frequently allows you to find an explicit equation where y is isolated cleanly on one side.

**3 Major Uses:**

• Population Dynamics: Predicting the growth rate of bacteria cultures or city populations over a decade.

• Chemical Mixing: Calculating the changing concentration of salt or chemicals in a fluid storage tank over time.

• Radioactive Decay: Determining how fast unstable nuclear isotopes lose mass over centuries.

**Real-Life Example:** Think of sorting coins. If you have a mixed pile of Rs 5 and Rs 10 coins, you separate them into two distinct piles before counting them. This method sorts variables into distinct piles before integrating them.

## Topic 4.3: Solution of Linear Differential Equations

**Definition:** A Linear Differential Equation is a structured equation where the dependent variable y and its derivative dy/dx appear only to the first power and are never multiplied together. They are solved using a special multiplying factor called an Integrating Factor (IF).

**3 Key Features:**

• Standard Form: It is written in a strict template format: dy/dx + Py = Q, where P and Q are either numbers or functions of x.

• Integrating Factor: You calculate a helper tool using the formula IF = e^(integral of P dx) to unlock the equation.

• Final Solution Structure: The ultimate solution follows a set blueprint: y * (IF) = integral of (Q * IF) dx + C.

**3 Major Uses:**

• Electrical Circuits: Calculating the exact current flowing through a circuit containing resistors and inductors (RL circuits).

• Fluid Mechanics: Tracking the velocity of an object falling through a thick fluid with constant drag resistance.

• Thermal Systems: Designing automated climate control systems that adjust heating based on external weather shifts.

**Real-Life Example:** Think of an airplane flying with a steady tailwind. The plane's forward acceleration is influenced by its engine power and the continuous push of the wind. A linear differential equation calculates the plane's exact velocity profile over time.